



Heal Smart

Smart Hospital Management System

Phase 3

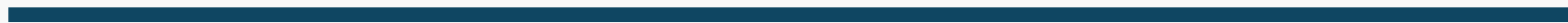


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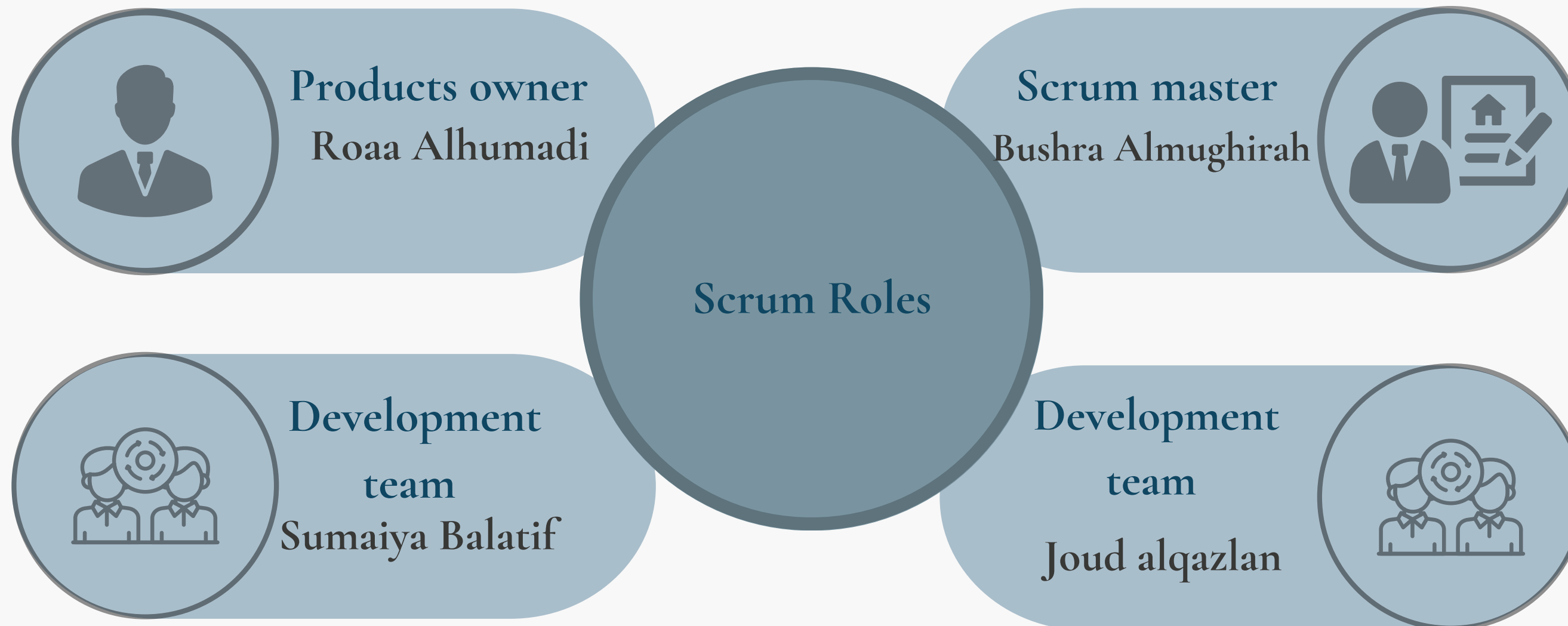
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Introdaction

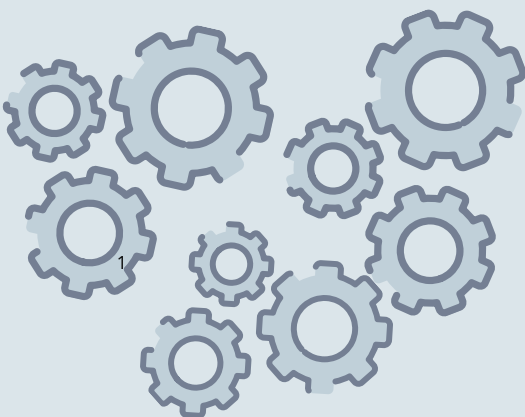
HealSmart is a comprehensive Hospital Management System that digitizes and integrates all healthcare operations into a unified platform. The system enables patients to book appointments, verify insurance, pay bills electronically, and access their medical records online. Healthcare providers can manage prescriptions, request lab tests, and view results in real-time, while administrative staff can handle employee management, generate financial reports, and coordinate with insurance companies. By replacing paper-based processes with digital solutions, HealSmart reduces medical errors, improves operational efficiency, enhances patient satisfaction, and .supports better healthcare delivery across the entire hospital ecosystem



Team Members



Final Product Backlog



PRODUCT BACKLOG							
ID	AS A ...	I WANT TO ...	SO THAT ...	Size	PRIORITY	Number of SPRINTs	STATUS
1	IT Administrator	register hospital staff	I can manage user access and system roles	S	High	1	Complete
2	Patient	register and create an account	I can book appointments and access my medical records	S	High	1	Complete
3	Patient	book an appointment online	so that I can choose a convenient time with my preferred doctor	L	High	1	Complete
3.1	Patient	view a list of available doctors	I can choose one based on my preferences	S	High	1	Complete
3.2	Patient	be able to select an available time slot for an appointment with the doctor	I can choose a time that suits me.	S	High	1	Complete
3.3	Patient	receive an email confirmation of my appointment	I have a record of it.	S	High	1	Complete
4	Doctor	view my patient list and appointments	I manage my daily schedule efficiently	M	High	1	Complete
5	Doctor	view a patient's full medical record before treatment	so that I can make accurate diagnoses and decisions	M	High	1	Complete
6	Patient	enter my insurance information in the hospital website	So that I can check if my insurance is valid and see my coverage amount	L	High	2	Complete
6.1	Patient	enter my ID number and insurance company name on the hospital's website	I can check if the insurance is valid to me or not.	S	High	2	Complete
6.2	Patient	receive a response through an interface that tells me whether my request is accepted or rejected	I can know if the insurance will cover my medical expenses.	S	High	2	Complete
6.3	Patient (note* if the insurance is valid)	the discount to be applied to the total amount due when paying the bill	I can benefit from the insurance coverage.	S	High	2	Complete
7	Lab Technician	upload test results to the system	doctors can access lab data without delay	S	High	2	Complete
8	hospital manager	be able to supervise and view all employee details and supported insurance companies in the hospital system	I can ensure smooth operations	S	Medium	2	Complete
9	Accountant Administrator	generate billing and payment reports	I can track hospital revenue and patient payments	S	Medium	2	Complete
10	Doctor	request lab tests and view results digitally	I can make faster medical decisions	S	Medium	2	Complete
11	Patient	pay medical bills online securely (visa , mada , apple pay)	I can complete my payments easily from home	S	Medium	2	Complete
12	Patient	view prescriptions and lab results online	I can stay informed about my health status	S	Medium	2	Complete
13	Patient	contact customer service via Chat	I can get assistance with my medical queries or issues	S	Medium	2	Complete
14	Patient	receive appointment reminders via email	I won't forget my scheduled visits	S	Low	1	Complete
15	Marketing Administrator	send promotional offers to patients via email	I can attract more customers	S	Low	2	Complete
TOTAL						33	

Sprint planing



Sprint Planning Report

Sprint: 2 **Date:** 22 Nov **Product:** Hospital Management

1. Sprint Goal:

- **Statement:** The goal of this sprint is to complete the planned features and improve the system's overall usability and efficiency.

2. Selected Product Backlog Items (with Acceptance Criteria):

- **User Story 1:**

- **Title:** ID and Insurance Company Validation

- **Description:** As a patient, I want to enter my ID number and insurance company name on the hospital's website so that I can check if my insurance is valid.

- **Acceptance Criteria:**

- The patient can enter their ID number and insurance company name.
 - The system performs validation of the entered data using an API.
 - The system notifies the patient whether their insurance is valid or invalid.

- **User Story 2:**

- **Title:** Insurance Response Status

- **Description:** As a patient, I want to receive a response from the insurance company through an interface that tells me whether my request is accepted or rejected, so I can know if my medical expenses will be covered.

User Stories and Tasks	ASSIGNED TO	ASSIGNED HOURS	HOURS USED	AVAILABLE HOURS REMAINING	DAY 1	DAY 2	DAY 3	DAY 4	DAY 5	DAY 6	DAY 7	DAY 8	DAY 9	DAY 10	DAY 11	DAY 12	DAY 13	DAY 14	DAY 15	DAY 16	DAY 17	DAY 18	DAY 19	Day:20	DAY 21
User Story 1 Name:As a patient i want to enter my ID number in insurance company name on the hospital's website, so that i can check if my insurance is valid to me or not.		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Task 1.1 ID and Insurance provider input screen	Member's Name:Joud	10	5	5	1	1	1	1	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Task 1.2 Insurance validation API integration	Member's Name: Sumaiya	12	6	6	2	1	1	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	
User Story 2 Name: As a patient I want receive a response from the insurance company through an interface that tells me whether my request is accepted or rejected so that i can know if the insurance will cover my medical expenses		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Task 2.1 Build response interface (accepted/rejected)	Member's Name: Roaa	14	6	8	0	0	2	2	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Task 2.2 Connect Ulwith insurance response API	Member's Name: Bushra	16	9	7	1	0	1	1	1	1	2	1	1	0	0	0	0	0	0	0	0	0	0	0	
User Story 3 Name: As a patient (note* if the insurance valid) i want to the discount to be applied to the total amount sue when paying the bill so that i can benefit from the insurance coverage		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Task 3.1 Discount calculation logic	Member's Name: Sumaiya	12	8	4	2	0	0	0	0	0	1	2	1	1	1	0	0	0	0	0	0	0	0	0	
Task 3.2 Apply discount in payment screen	Member's Name: Joud	10	6	4	1	2	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
User Story 4 Name: As a lab technician i want to uoload test result to the system so that doctor can access lab data without delay.		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Task 4.1 Uplod module (UI+file validation)	Member's Name: Bushra	14	7	7	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	
Task 4.2 Backebd integration to store lnh results	Member's Name: Sumaiya	16	9	7	0	0	0	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	
User Story 5 Name:As a hospital manager i want to be able to service and view all employee details and supported insurance companies in the hospital system so that i can ensure smooth operations		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Task 5.1 Employee management screen	Member's Name: Roaa	12	8	4	0	0	0	1	1	1	1	1	2	1	0	0	0	0	0	0	0	0	0	0	
Task 5.2 Insurance company managaement module	Member's Name: Bushra	14	6	8	0	0	0	0	0	0	0	0	1	1	1	1	1	1	0	0	0	0	0	0	
User Story 6 Name: As a Accountant administrator i want to generate billing and payment reports so that i cant track hospital revenue and patient payments		0	0	0	0	0	0	0	0	0	0	0	0	P	0	0	0	0	0	0	0	0	0	0	
Task 6.1 Billing report generator	Member's Name: Joud	10	6	4	0	0	0	0	0	0	0	0	0	0	0	2	2	1	1	0	0	0	0	0	
Task 6.2 Payment report and export online	Member's Name: Bushra	12	5	7	0	1	1	2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
User Story 7 Name:As a doctor i want to request lab test and view result digitlly so that is can make faster medical decision		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Task 7.1 Lab test request form	Member's Name: Roaa	8	4	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	0	
Task 7.2 Link test results to patient record	Member's Name: Sumaiya	14	8	6	1	2	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	
User Story 8 Name:As a patient i want to pay medical bills online securely (visa, mada, apple pay), so that is can complete my payments easily from home.		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Task 8.1 Payment gateway integration	Member's Name: Sumaiya	16	8	8	1	1	2	1	2	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	
Task 8.2 Payment receiptand confirmation	Member's Name: Joud	10	6	4	0	0	0	0	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	
User Story 9 Name:As a patient i want to view prescription and lab results online so that i can stay informed about my health status		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Task 9.1 Proscription biewing page	Member's Name: Roaa	8	8	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	0	
Task 9.2 Link lab results to patient portal	Member's Name: Bushra	12	6	6	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	0	
User Story 10 Name:As a patient i want to contact customer service via chat so that i can get assistance with my medical queries or issues.		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Task 10.1 chat UI	Member's Name: Joud	10	6	4	0	0	2	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	
Task 10.2 Backend chat service connection	Member's Name: Sumaiya	14	8	6	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	
User Story 11 Name :As a marketing administrator i want to send promotional offers to patients via email so that i can attract more customers		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Task 11.1 Email template and UI	Member's Name: Bushra	8	5	3	2	1	1	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	0	0	
Task 11.2 Email sending service integration	Member's Name: Sumaiya	12	10	2	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	0	
TOTAL		264	150	114																					

Scrum Metting

Week 5 :

DAILY SCRUM						
Team member	Question	Sunday	Monday	Tuesday	Wednesday	Thursday
Joud	What did you do yesterday?	Plan Sprint 2	ID and Insurance provider screen	Continue insurance screen	Insurance validation API integration	Test insurance validation
	What are doing today?	ID and Insurance provider screen	Continue insurance screen	Insurance validation API integration	Test insurance validation	Review insurance module
	Is there anything blocking you?	No	No	No	API issues	No
Roaa	What did you do yesterday?	Review Sprint 2 backlog	Build response interface accepted/rejected	Continue response interface	Connect UI with insurance response API	Test response interface
	What are doing today?	Build response interface accepted/rejected	Continue response interface	Connect UI with insurance response API	Test response interface	Employee management screen
	Is there anything blocking you?	No	No	No	No	No
Sumaiya	What did you do yesterday?	Select Sprint 2 tasks	Insurance validation API integration	Continue API work	Discount calculation logic	Apply discount in payment screen
	What are doing today?	Insurance validation API integration	Continue API work	Discount calculation logic	Apply discount in payment screen	Test discount
	Is there anything blocking you?	No	No	Connection timeout	No	No
Bushra	What did you do yesterday?	Assign tasks	Build lab results upload interface	Continue Building lab results upload interface	Backend integration to store lab results	Test upload
	What are doing today?	Build lab results upload interface	Continue Building lab results upload interface	Backend integration to store lab results	Test upload	Lab test request form
	Is there anything blocking you?	No	No	No	No	No



Scrum Metting

Week 6 :

			DAILY SCRUM			
Team member	Question	Sunday	Monday	Tuesday	Wednesday	Thursday
Joud	What did you do yesterday?	Review insurance module	Payment gateway integration	Continue payment integration	Payment receipt and confirmation	Test payment
	What are doing today?	Payment gateway integration	Continue payment integration	Payment receipt and confirmation	Test payment	Prescription viewing page
	Is there anything blocking you?	No	No	No	Payment sandbox issues	No
Roaa	What did you do yesterday?	Employee management screen	Continue employee screen	Insurance company management module	Test employee management	Chat UI
	What are doing today?	Continue employee screen	Insurance company management module	Test employee management	Chat UI	Continue chat interface
	Is there anything blocking you?	No	No	No	No	No
Sumaiya	What did you do yesterday?	Test discount	Billing report generator	Continue report generator	Payment report and export options	Test reports
	What are doing today?	Billing report generator	Continue report generator	Payment report and export options	Test reports	Email template and UI
	Is there anything blocking you?	No	No	No	No	No
Bushra	What did you do yesterday?	Lab test request form	Continue lab request	Link test results to patient record	Test lab module	Backend chat service connection
	What are doing today?	Continue lab request	Link test results to patient record	Test lab module	Backend chat service connection	Email sending service integration
	Is there anything blocking you?	No	No	No	No	No



Scrum Metting

Week 7 :

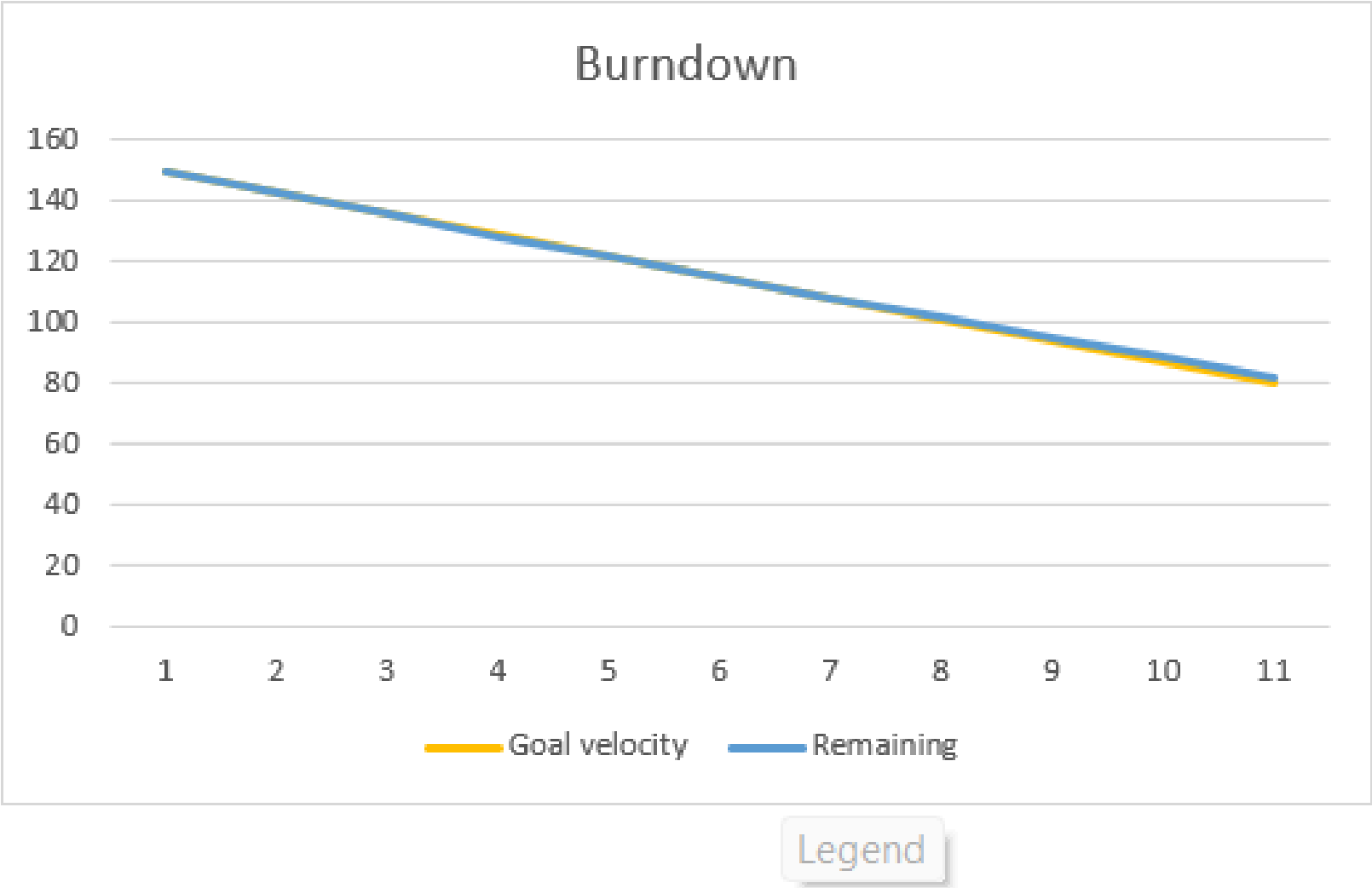
			DAILY SCRUM			
Team member	Question	Sunday	Monday	Tuesday	Wednesday	Thursday
Joud	What did you do yesterday?	Prescription viewing page	Continue prescription page	Link lab results to patient portal	Test prescription and lab results	Review all my work
	What are doing today?	Continue prescription page	Link lab results to patient portal	Test prescription and lab results	Review all my work	Prepare final demo
	Is there anything blocking you?	No	No	No	No	No
Roaa	What did you do yesterday?	Continue chat interface	Test chat UI	Backend chat service connection	Test chat service	Review all features
	What are doing today?	Test chat UI	Backend chat service connection	Test chat service	Review all features	Final testing
	Is there anything blocking you?	No	No	No	No	No
Sumaiya	What did you do yesterday?	Email template and UI	Continue email template	Email sending service integration	Test email service	Review billing and reports
	What are doing today?	Continue email template	Email sending service integration	Test email service	Review billing and reports	Sprint 2 review
	Is there anything blocking you?	No	No	No	No	No
Bushra	What did you do yesterday?	Email sending service integration	Test email integration	Review all backend work	System testing	Prepare presentation
	What are doing today?	Test email integration	Review all backend work	System testing	Prepare presentation	Sprint 2 retrospective
	Is there anything blocking you?	No	No	No	No	No



Burdown Chart

BURNDOWN CHART SPRINT2

Days	Goal	Done	Goal velocity	Remaining
0	0	0	150	150
1	7	7	143	143
2	7	7	136	136
3	7	8	129	128
4	7	6	122	122
5	7	7	115	115
6	7	7	108	108
7	7	6	101	102
8	7	7	94	95
9	7	6	87	89
10	7	7	80	82
11	7	7	73	75
12	7	8	66	67
13	7	7	59	60
14	7	9	52	51
15	7	8	45	43
16	7	8	38	35
17	7	7	31	28
18	7	8	24	20
19	7	6	17	14
20	8	7	9	7
21	9	7	0	0



Sprint review report

Sprint Review Report

Sprint: [Sprint 2] **Date:** [23 November 2025] **Product:** [Hospital Management System-HealSmart]

I. Sprint Goal Achievement and Completed Work:

- **1.1 Sprint Goal Review:**

- **Sprint Goal Achieved:** [Yes]
- **Achievement Details:** The development team successfully completed all user stories with their associated tasks. All core functionalities have been implemented and including: Insurance verification system, Payment processing with multiple methods, Lab results management, Employee and insurance company management

1.2 Completed Work Demonstration:

- The Development Team demonstrated the following completed Product Backlog items:
- 1. Insurance verification
- 2. Insurance response interface
- 3. Automatic discount application,
- 4. Lab results upload system
- 5. Employee and insurance management
- 6. Billing and payment reports
- 7. Digital lab test requests
- 8. Secure online payment
- 9. Patient portal for prescriptions and lab results
- 10. . Email marketing campaigns
- 11.costumer service chat

Sprint Retrospective

Date: 20nov

Meeting facilitator:Roaa

Start

- Code review sessions

Schedule dedicated time for peer code reviews to improve code quality and share knowledge. This practice will help catch issues early and promote best practices across the team

- Automated testing

Begin implementing unit tests and integration tests for critical features to catch bugs early and improve code reliability.

- Sprint planning poker

Use estimation techniques like planning poker to improve our task estimation accuracy and reach team consensus on effort required.

Stop

- Multitasking across multiple user stories

Stop working on multiple user stories simultaneously. Focus on completing one story before moving to the next to improve flow and reduce context switching.

- Waiting for perfect solutions

Stop over-engineering solutions or waiting for the perfect approach. Embrace iterative development and continuous improvement.

Continue

- Strong team collaboration

Continue the excellent collaboration . The team showed great willingness to help each other and share knowledge throughout the sprint.

- Meeting sprint commitments

Continue delivering on our sprint commitments. We completed all 11 user stories as planned, which demonstrates our ability to plan and execute effectively.

- Clear task breakdown

Continue breaking down user stories into manageable tasks. The task decomposition in this sprint was well-structured and helped track progress effectively.



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Thank You

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